

Vasoten F.C. Tablets 5mg

(Bisoprolol Fumarate)

17/08/2010

Composition:

Each tablet contains: Bisoprolol Fumarate 5mg

Product Description:

Yellow to light-yellow, heart-shaped, biconvex film coated tablets, and vertically scored in half on both sides.

Clinical Pharmacology:

- Pharmacodynamics:

Bisoprolol is a beta1-selective adrenergic blocking agent. The drug does not exhibit the intrinsic sympathomimetic activity seen with pindolol or the membrane-stabilizing activity possessed by propranolol or pindolol. At low dosages, bisoprolol selectively inhibits response to adrenergic stimuli by competitively blocking cardiac beta1-adrenergic receptors, while having little effect on the beta2-adrenergic receptors of bronchial and vascular smooth muscle. At high doses (e.g., 20mg or higher), the selectivity of bisoprolol on beta1- and beta2-adrenergic receptors usually diminishes, and the drug will competitively inhibit and adrenergic receptors.

- Pharmacokinetics:

Bisoprolol is absorbed almost completely from the gastrointestinal tract. Together with the very small first pass effect in the liver, this results in a high bioavailability of approximately 90%. The drug is cleared equally by the liver and kidneys.

The long plasma half-life (9-12 hours) provides 24-hour efficacy following a once daily dosage. About 95% of the drug substance is excreted through the kidneys, 50% of a dose as unchanged bisoprolol. There are no active metabolites in man.

There are limited data in children and an appropriate dosing regimen has not been established. Therefore, bisoprolol fumarate cannot be recommended for use in children.

The pharmacokinetics of bisoprolol fumarate in the pediatric population were examined in three independent studies and showed that the half-life of bisoprolol fumarate ranged from 3.6 hours to 14 hours and the clearance ranged from 0.12 L/hr/kg to 0.44 L/hr/kg. In adults, the half-life of bisoprolol fumarate ranged from 9 hours to 12 hours and the clearance ranged from 0.14 L/hr/kg to 0.15 L/hr/kg.

Indication:

The management of hypertension and angina pectoris.

Dosage and Administration:

Treatment of hypertension or angina pectoris

In all cases the dose regimen is adjusted individually by your doctor, in particular according to the pulse rate and therapeutic success.

The usual initial dose is 5 mg bisoprolol (1 tablet of Vasoten 5 mg) once daily. If necessary, the dose may be increased to 10 mg bisoprolol (2 tablets of Vasoten 5 mg) once daily.

The maximum recommended dose is 20 mg bisoprolol once daily.

Treatment of stable chronic heart failure

The initiation of treatment of stable chronic heart failure with Vasoten necessitates a special titration phase and requires regular monitoring by the doctor.

Preconditions for treatment with bisoprolol are:

- stable chronic heart failure without acute failure during the past six weeks,
- mainly unchanged basic therapy during the past two weeks,
- treatment at optimal dose with an ACE inhibitor (or other vasodilator in case of intolerance to ACE inhibitors) and a diuretic, and optionally cardiac glycosides.

It is recommended that the treating physician be experienced in the management of chronic heart failure.

The treatment of stable chronic heart failure with bisoprolol is initiated according to the following titration scheme, individual adaptation may be necessary depending on how well the patient tolerates each dose, i.e. the dose is to be increased only, if the previous dose is well tolerated.

1 st week:	1.25 mg bisoprolol once daily*
2 nd week:	2.5 mg bisoprolol (½ tablet Vasoten 5 mg) once daily
3 rd week:	3.75 mg bisoprolol once daily*
4 th – 7 th week:	5 mg bisoprolol (1 tablet Vasoten 5 mg) once daily
8 th – 11 th week:	7.5 mg bisoprolol (1½ tablets Vasoten 5 mg) once daily
12 th week and beyond:	10 mg bisoprolol (2 tablets Vasoten 5 mg) once daily as maintenance treatment

* Vasoten 5 mg is not suitable for initial treatment of stable chronic heart failure. Lower strengths are available for this purpose.

The maximum recommended dose is 10 mg bisoprolol once daily. Patients should be titrated to and maintained at this dose unless prevented by adverse effects.

After initiation of treatment with 1.25 mg bisoprolol, the patient should be observed over a period of approximately 4 hours (especially as regards blood pressure, heart rate, conduction disturbances, signs of worsening of heart failure).

During the titration phase or thereafter, transient worsening of heart failure, fluid retention, hypotension or bradycardia may occur. In this case it is recommended first to reduce the dose of bisoprolol. Bisoprolol should be discontinued only if clearly necessary, but its reintroduction and/or uptitration should always be considered when the patient becomes stable again.

Duration of treatment for all indications

Treatment with Vasoten 5 mg is generally a long-term therapy.

The treatment may be interrupted if necessary and reintroduced as appropriate.

Do not stop treatment abruptly or change the recommended dose without talking to your doctor first since this might lead to a transitory worsening of heart condition. Especially in patients with ischaemic heart disease, treatment must not be discontinued suddenly. If discontinuation is necessary, the dose is gradually decreased.

Special populations

Renal or hepatic impairment:

- *Treatment of hypertension or angina pectoris:* In patients with liver or kidney function disorders of mild to moderate severity no dosage adjustment is normally required. In patients with severe renal impairment (creatinine clearance < 20 ml/min) and in patients with severe hepatic impairment a daily dose of 10mg bisoprolol must not be exceeded.

- *Treatment of stable chronic heart failure:* There is no information regarding pharmacokinetics of bisoprolol in patients with chronic heart failure and concomitant hepatic or renal impairment. Titration of the dose in these populations must therefore be made with particular caution.

Elderly

No dosage adjustment is required.

Administration

Vasoten 5 mg tablets are taken in the morning with or without food. They are swallowed with some liquid and not to be chewed.

Contraindication:

Patients with:

- Untreated cardiac failure
- Cardiogenic shock
- Sinoatrial block
- Second or third degree AV block
- Marked bradycardia
- Severe asthma
- Sick sinus syndrome
- Hypotension
- Severe peripheral circulatory disturbances
- Hypersensitivity to bisoprolol in Vasoten
- Untreated tumours of the adrenal gland (phaeochromocytoma)
- Metabolic acidosis

Warning and Precaution:

The following section describes when Vasoten must be used with special caution:

- diabetes mellitus with extremely fluctuating blood glucose levels: symptoms of markedly reduced blood glucose (hypoglycaemia) such as tachycardia, palpitations or sweating can be masked,
- strict fasting,
- ongoing desensitisation therapy,
- mild disturbances of atrioventricular conduction (first degree AV block),
- disturbed blood flow in the coronary vessels due to vasospasms (Prinzmetal's angina),
- peripheral arterial occlusive disease (intensification of complaints may occur especially when starting therapy)
- patients with a personal or family history of psoriasis.

Respiratory system:

Beta-blockers should generally be avoided in patients with a history of asthma or chronic obstructive airway disease. However, if there is no alternative, a cardio-selective beta-blocker such as Vasoten may be used with extreme caution under specialist supervision. In some asthmatic patients, some increase in airway resistance may occur, and this may be regarded as a signal to discontinue therapy. Bronchospasm can usually

be reversed by commonly used bronchodilators such as salbutamol. As a warning to patients, both the label and patient information leaflet for Vasoten advise patients not to take Vasoten, but to talk to their doctor if they have ever suffered from wheezing or asthma.

Allergic reactions:

Beta-blockers, including Vasoten, may increase the sensitivity to allergens and the severity of anaphylactic reactions because the adrenergic counterregulation under beta-blockade may be alleviated. Treatment with adrenaline may not always give the expected therapeutic effect.

General anaesthesia:

In patients undergoing general anaesthesia the anaesthetist must be aware of beta-blockade. If it is thought necessary to withdraw Vasoten before surgery, this should be done gradually and completed about 48 hours prior to anaesthesia.

Phaeochromocytoma:

In patients with a tumour of the adrenal gland (phaeochromocytoma) Vasoten may only be administered after previous alpha-receptor blockade.

Thyrotoxicosis: Under treatment with Vasoten the symptoms of a thyroid hyperfunction (thyrotoxicosis) may be masked.

Bisoprolol should be used with care in patients with a prolonged PR conduction interval, poor cardiac reserve and peripheral circulatory disease such as Raynaud's phenomena, since aggravation of these disorders may occur. Treatment should not be discontinued abruptly. As with other beta-blockers, gradual dosage reduction over 1-2 weeks is recommended, particularly in patients with ischaemic heart disease. If necessary, replacement therapy should be initiated at the same time to prevent exacerbation of angina pectoris

In the event of a precipitous drop in pulse rate and/or blood pressure, treatment with Vasoten should be discontinued.

Bisoprolol should be used with caution in patients with metabolic acidosis.

Due to their negative effect on conduction time, beta-blockers should only be given with caution to patients with first degree heart block.

Caution should be exercised when using anaesthetic agents with Vasoten. The anaesthetist should be informed if the patient is taking Vasoten, and anaesthetic agents causing myocardial depression, such as cyclopropane or trichloroethylene, are best avoided. In cases of severe ischaemic heart disease, the risk/benefit of continuing treatment should be carefully evaluated. If withdrawal of Vasoten is desired, this should be completed at least 24 hours prior to anaesthesia. Continuation of beta-blockade reduces the risk of arrhythmias during induction and intubation, but may result in attenuation of reflex tachycardia and increase the risk of hypotension. The patient may be protected against vagal reactions by intravenous administration of atropine.

Use in pregnancy and lactation

During pregnancy Vasoten should only be recommended following careful assessment of benefit-to-risk ratio by the doctor. In general, beta-blockers reduce placental blood flow and may affect the development of the unborn child. Placental and uterine blood flow as well as the

growth of the unborn child must be monitored and, in case of harmful effects on pregnancy or the foetus, alternative therapeutic measures considered.

The newborn infant must be monitored closely after delivery. Symptoms of reduced blood glucose and slowed pulse rate generally may occur within the first 3 days of life.

There are no data on the excretion of bisoprolol in human breast milk or the safety of bisoprolol exposure in infants. Therefore administration of Vasoten is not recommended during breastfeeding.

Side Effects:

The following definitions apply to the frequency terminology used hereafter:

Very common ($\geq 1/10$)

Common ($\geq 1/100, < 1/10$)

Uncommon ($\geq 1/1,000, < 1/100$)

Rare ($\geq 1/10,000, < 1/1,000$)

Very rare ($< 1/10,000$)

Cardiac disorders:

Very common: bradycardia.

Common: worsening of heart failure

Uncommon: AV-conduction disturbances.

Very rare: chest pain.

Investigations:

Rare: increased triglycerides, increased liver enzymes (ALAT, ASAT).

Nervous system disorders:

Common: dizziness, headache.

Eye disorders:

Rare: reduced tear flow (to be considered if the patient uses lenses).

Very rare: conjunctivitis.

Ear and labyrinth disorders:

Rare: hearing impairment.

Respiratory, thoracic and mediastinal disorders:

Uncommon: bronchospasm in patients with bronchial asthma or a history of obstructive airways disease.

Rare: allergic rhinitis.

Gastrointestinal disorders:

Common: gastrointestinal complaints such as nausea, vomiting, diarrhoea, constipation.

Skin and subcutaneous tissue disorders:

Rare: hypersensitivity reactions (itching, flush, rash).

Very rare: beta-blocking agents may provoke or worsen psoriasis or induce psoriasis-like rash, alopecia.

Musculoskeletal and connective tissue disorders:

Uncommon: muscular weakness and cramps.

Vascular disorders:

Common: feeling of coldness or numbness in the extremities, hypotension.

Uncommon: orthostatic hypotension.

General disorders:

Common: asthenia, fatigue.

Hepatobiliary disorders:

Rare: hepatitis.

Reproductive system and breast disorders:

Rare: potency disorders.

Psychiatric disorders:

Uncommon: sleep disorders, depression.

Rare: nightmares, hallucinations.

Overdose:

- If overdose occurs, bisoprolol treatment should be stopped. Supportive therapy and symptomatic treatment should be provided. Limited data suggest bisoprolol fumarate cannot be dialysed.
- Symptoms of overdose with a beta-blocker may include bradycardia, hypotension, bronchospasm and acute cardiac insufficiency. Bradycardia or extensive vagal reactions may be countered by atropine.
- Hypotension and shock may be treated with intravenous fluids, and, if necessary, catecholamines. The beta-blocking effect may be counteracted by slow intravenous administration of isoprenaline or dobutamine. In refractory cases, isoprenaline may be combined with dopamine. If this does not produce the desired effect, intravenous glucagon may be considered. The administration of calcium ions or the use of a cardiac pacemaker may also be considered.
- AV block (second or third degree): Patients should be carefully monitored and treated with isoprenaline or cardiac pacemaker insertion, as appropriate.
- Acute worsening of heart failure: Administration of intravenous diuretics and inotropic agents should be considered.
- Bronchospasm can usually be reversed by intravenous bronchodilators.

Drug Interaction:

The effect and tolerability of medicines can be influenced by simultaneous intake of other medication. Such interactions can also occur if a short time has elapsed since the use of the other medication. Tell your doctor if you are taking any other medicine - even those not prescribed to you by a doctor.

Combinations not recommended

Treatment of stable chronic heart failure

Class-I antiarrhythmic medicines [e.g. quinidine, disopyramide, lidocaine, phenytoin; flecainide, propafenone) may increase the depressant effect of Vasoten on atrio- ventricular impulse conduction and the contractility of the heart.

All indications

Calcium antagonists of the verapamil type and to a lesser extent of the diltiazem type may lead to reduced contractility of the heart muscle and

delayed atrio-ventricular impulse conduction when used concomitantly with Vasoten. Especially intravenous administration of verapamil in patients on β -blocker treatment may lead to profound hypotension and atrioventricular block.

Centrally acting blood pressure-lowering medicines [such as clonidine, methyldopa, moxonidine, rilmenidine) may lead to a reduction of heart rate and cardiac output, as well as to vasodilation due to a decrease in the central sympathetic tonus. Abrupt withdrawal, particularly if prior to beta-blocker discontinuation, may increase risk of “rebound hypertension”.

Combinations to be used with caution

Treatment of hypertension or coronary heart disease (angina pectoris)

Class-I antiarrhythmic medicines [e.g. quinidine, disopyramide, lidocaine, phenytoin; flecainide, propafenone) may increase the depressant effect of Vasoten on atrio- ventricular impulse conduction and the contractility of the heart.

All indications

Calcium antagonists of the dihydropyridine type (e.g. nifedipine) may increase the risk of hypotension when used concomitantly with Vasoten. An increased risk of a further deterioration of the ventricular pump function in patients with heart failure cannot be excluded.

Class-III antiarrhythmic medicines (e.g. amiodarone) may increase the inhibitory effect of Vasoten on atrio-ventricular impulse conduction.

Topical β -blockers (e.g. eye drops for glaucoma treatment) may add to the systemic effects of Vasoten.

Parasympathomimetic medicines may increase the inhibitory effect on atrio-ventricular impulse conduction and the risk of bradycardia when used concomitantly with Vasoten.

The blood sugar lowering effect of insulin or oral antidiabetic medicines may be intensified. Warning signs of reduced blood glucose (hypoglycaemia) - especially accelerated heart rate (tachycardia)- may be masked or suppressed. Such interactions are considered to be more likely with nonselective β -blockers.

Anaesthetic agents may increase the risk of cardiodepressive actions of Vasoten, leading to hypotension.

Cardiac glycosides (digitalis) may lead to an increase in impulse conduction time and thus reduction in heart rate when used concomitantly with Vasoten.

Non-steroidal anti-inflammatory medicines (NSAIDs) may reduce the blood pressure- lowering effect of Vasoten.

β -Sympathomimetics (e.g. isoprenaline, dobutamine) used in combination with Vasoten may lead to a reduced effect of both agents.

A combination of Vasoten with sympathomimetics that activate both β - and α -adrenoceptors [e.g. noradrenaline, adrenaline) may intensify the α -adrenoceptor-mediated vasoconstrictor effects of these agents leading to blood pressure increase and exacerbated intermittent claudication. Such interactions are considered to be more likely with nonselective β -blockers.

Antihypertensive agents as well as other medicines with blood pressure lowering potential (e.g. tricyclic antidepressants, barbiturates, phenothiazines) may increase the blood pressure lowering effect of Vasoten.

Combinations to be considered

Mefloquine may increase the risk of decelerating the heart rate (bradycardia), if used in combination with Vasoten.

Monoamine oxidase inhibitors (except MAO-B inhibitors) may enhance the hypotensive effect of the beta-blockers. Concomitant use may also be a risk for hypertensive crisis.

Storage:

Store below 30°C and keep medicine out of reach of children.

Vasoten tablets should be stored in the original container and should not be removed from their foil pack until required for administration.

Pack Size:

10x10 tablets per box

Manufacturer:

Synmosa Biopharma Corporation, Synmosa Plant

No.6, Kuang Yeh 1st Road, Hsin-Chu Ind. Park, Hsin Chu Hsien, Taiwan, R.O.C.